



Research paper

SINDy-based modeling of an uncrewed surface vessel with experimental validation

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ABSTRACT

This paper introduces physics-based digital twins of uncrewed surface vehicles (USVs) using the Sparse Identification of Nonlinear Dynamical Systems (SINDy) framework. The identified governing dynamics are embedded into the digital twins to enable accurate simulation of USV behavior. Field experiments involving a range of maneuvers, both linear and curved trajectories, are conducted to collect operational data. To streamline the modeling process, the vehicle dynamics are decomposed into longitudinal and lateral subsystems, considering three degrees of freedom (DOF). Data from the onboard central computer are used to identify the system's nonlinear dynamics matrix. Several candidate function libraries within the SINDy framework, including first- and second-degree polynomials and a physics-informed custom library, are tested for model training and validation. Root mean square error (RMSE) is employed as the performance metric. Results show that the proposed custom library captures the vehicle dynamics more accurately, achieving a 26.8% lower average RMSE with reduced standard deviation compared to the polynomial-based models.

1. Introduction

Uncrewed surface vehicles (USVs) are increasingly used in marine applications such as environmental monitoring, offshore inspection, and data collection in areas that are difficult or unsafe for human access. The safe and efficient operation of these vehicles depends on accurate models that can support navigation, control, and prediction in uncertain marine environments. Developing such models is challenging. Traditional physics-based models often rely on simplifying assumptions that limit their ability to capture the nonlinear hydrodynamic effects and complex interactions with waves, currents, and wind. On the other hand, purely data-driven models can suffer from overfitting and sensitivity to noise, especially when measurements are affected by environmental disturbances and sensor limitations. As a result, existing approaches do not provide the balance of accuracy, robustness, and interpretability needed for reliable deployment of autonomous USVs.

To address this gap, the present work is based in part on material from the author's thesis titled "Sparse Identification of Nonlinear Dynamics in Digital Twins for an Unmanned Surface Vessel" submitted to Norwegian University of Science and Technology in 2023 (Kandemir, 2023), and investigates the use of system identification techniques for building dynamic models directly from experimental data. In particular, it applies the Sparse Identification of Nonlinear Dynamics (SINDy) framework (Brunton et al., 2016), which is designed to extract governing equations from noisy measurements while promoting model

simplicity. Integrating such a model into a physics-based digital twin of the Otter USV enables real-time simulation and validation against experimental trials. The goal is to develop a modeling approach that better represents USV dynamics under realistic conditions while remaining computationally efficient for control and optimization tasks. The expected outcomes include supporting future developments in digital twins for marine systems and providing practical benefits for the safe and reliable use of USVs in research, industry, and environmental applications.

1.1. Literature review

The following section reviews related work in USV modeling and system identification to highlight current approaches and their limitations. One of the widely used methods in system identification is least squares regression. This method can provide accurate dynamic models if the data is linear in parameters and subject to Gaussian white noise (Ljung, 2012). However, in the real world, the noise in the measured data is biased and subject to colored (correlated) noise, which yields substantial uncertainties. Several related works in the literature have considered uncertainties in system identification of surface vessel dynamics. Two methods have been employed by Xu et al. (2019): singular value decomposition and Tikhonov regularization. The R^2 goodness of fit criterion demonstrates the model's accuracy. In a

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study by [Sutulo and Guedes Soares \(2014\)](#), a genetic algorithm based method is applied to artificially generated data with a high amount of white noise. The study aims to minimize a cost function between reference data and the identified parameters. In a paper written by Hinostroza, [\(Xu et al., 2018\)](#), similar research was conducted to eliminate the uncertainty using a global optimization algorithm with various loss functions. The method is called a modified version of the Abkowitz model. However, comparing to [Sutulo and Guedes Soares \(2014\)](#), the data are collected from a vessel in an experiment. In another paper written [\(Miller, 2021\)](#) genetic algorithm is used to optimize bias and variance in the regularization process. However, the model used for the vessel is linearized, so the model identified cannot represent the complete characteristics of the system. In another paper by [Perera et al. \(2015\)](#), a stochastic method is proposed to identify the parameters for vessel steering. The method uses a modified version of the Nomoto model for navigation and extended Kalman filter to estimate the parameters. Although extended Kalman filter is an effective algorithm, it does not guarantee convergence for all situations and fails to achieve the expected accuracy [\(Ungarala et al., 2007\)](#). Extended Kalman filter has been used for ship maneuvering system identification in another study. In the study by [Alexandersson et al. \(2022\)](#), the extended Kalman filter is accompanied by inverse dynamics regression. Instead of feeding the raw data into the model, Rauch–Tung–Striebel smoother is applied for clean data. Similar to [Xu et al. \(2018\)](#), the experiment data is used rather than the synthetic one. Support vector machine is another method that is implemented for parameter identification. The study presented by [Suykens and Vandewalle \(1999\)](#) proposes a least squares version of the support vector machine. The formulation enables a solution for a set of linear equations instead of traditional quadratic programming in support vector machine. Although the study is formulated for binary classification, its formulation is later adapted to parameter estimation in the subsequent studies.

The study by [Luo and Zou \(2009\)](#) estimates nonlinear maneuvering model coefficients with a support vector machine. A batch algorithm is used without initial estimations. For verification purposes, 20/20 degree zigzag maneuver is adapted. The support vector machine is sensitive to the regularization and kernel parameters. Moreover, the analytical calculation of the optimal parameter is challenging [\(Vapnik et al., 1994\)](#). Therefore, in the study by [Luo et al. \(2016\)](#), particle swarm optimization is used to determine these parameters. Another alternative for the structural parameter optimization in support vector machine is the artificial bee colony algorithm applied in the study by [Zhu et al. \(2017\)](#). Alternatively, the optimal truncated least squares support vector machine method can be used for kernel matrix reduction while eliminating the several matrix inversion in singular value decomposition [\(Xu et al., 2020\)](#). With the advances in computational power, the neural network can be an alternative method to identify the vessel's dynamic characteristics. In the study by [He et al. \(2022\)](#), neural network modeling for vessel dynamics is used with some experimental data. Although precise results are obtained, the model interpretability is low. Recent work [\(Liu et al., 2024\)](#) has demonstrated the potential of SINDy for scaled vehicle models in laboratory conditions, but the robustness of the approach under real-world disturbances and with full-scale vessels remains unexplored.

Accurate system identification of surface vessel dynamics remains a challenging task due to the nonlinear behavior of vessels and the presence of disturbances and measurement uncertainties in real-world environments. Existing methods such as least squares regression, extended Kalman filtering, support vector machines, and neural networks have shown promising results, but they are often limited by assumptions of linearity, sensitivity to noise, high computational demand, or lack of interpretability. Moreover, many prior studies rely on synthetic data or scaled-down experimental setups under controlled conditions, which do not fully capture the complexities encountered by full-scale USVs operating offshore.

1.2. Contribution

This paper addresses the challenges of system identification for uncrewed surface vehicles (USVs) in real-world marine environments, where existing methods face limitations of linear assumptions, sensitivity to noise, reliance on synthetic data, and lack of interpretability. The main contributions are as follows:

1. **Full-scale experimental validation of SINDy for USVs:** Unlike many prior studies that use synthetic or laboratory-scale datasets, this work applies the Sparse Identification of Nonlinear Dynamics (SINDy) framework to experimental data collected from a full-scale Otter USV in offshore conditions. This provides new evidence of SINDy's applicability under realistic disturbances and measurement uncertainties.
2. **Development of an interpretable and scalable dynamic model:** A custom library of candidate functions is designed to capture relevant nonlinear hydrodynamic effects while maintaining model interpretability. This adjustment improves both accuracy and robustness across diverse maneuvers, while keeping the model compact enough for efficient use in simulation and control.
3. **Integration into a physics-based digital twin framework:** The identified model is embedded in a digital twin environment, enabling repeatable simulation, performance analysis, and comparison with real-world trials. This establishes a foundation for future applications in mission planning, fault detection, and lifecycle management of USVs.

Together, these contributions fill the gap in applying interpretable, data-driven system identification to full-scale USVs under realistic operating conditions, advancing the development of reliable digital twins for marine systems.

2. Methodology

2.1. Experimental setup

An experimental setup was executed to capture the characteristics of both the longitudinal and lateral motion to perform system identification of the USV. The objective was to adequately excite the system and identify the dynamics associated with each motion component. A series of maneuvers were conducted in the longitudinal motion, followed by maneuvers to target the lateral motion which are shown in [Fig. 1](#) on a map. The coordinates of the maneuvers shown on the map are extracted as longitudinal and latitude values from the sensors on board. The experiment involved performing straight-line maneuvers at various velocities for longitudinal motion identification which are shown between [Figs. 1\(a\)–1\(d\)](#). By conducting these maneuvers, the USV was subjected to different velocities in various linear paths, allowing for observing its longitudinal dynamics. This enabled the collection of data that would assist in estimating the parameters associated with the longitudinal motion. Similarly, to examine the lateral motion of the USV, circular maneuvers were employed which are shown between [Figs. 1\(e\)–1\(h\)](#). These circular trajectories were executed at different radii and velocities, providing a comprehensive range of lateral motion data. By performing circular maneuvers, the USV experienced different thrust and torque inputs, allowing for the identification of the associated parameters. The acquired data facilitated estimating the USV's behavior in lateral motion. The maneuvers are carefully planned to induce the desired longitudinal and lateral motion responses. This experimental setup aimed to capture the complete dynamics of the USV. The resulting data serve as the basis for system identification and model prediction, contributing to a better understanding of the USV's behavior in different operational scenarios.

Using different libraries is part of identifying dynamic models. Three main candidate functions, the first order polynomial, the second order polynomial, and a specially designed library, have been used in the current analysis. [Table 1](#) offers the necessary information to understand possible combinations of states and input vectors.

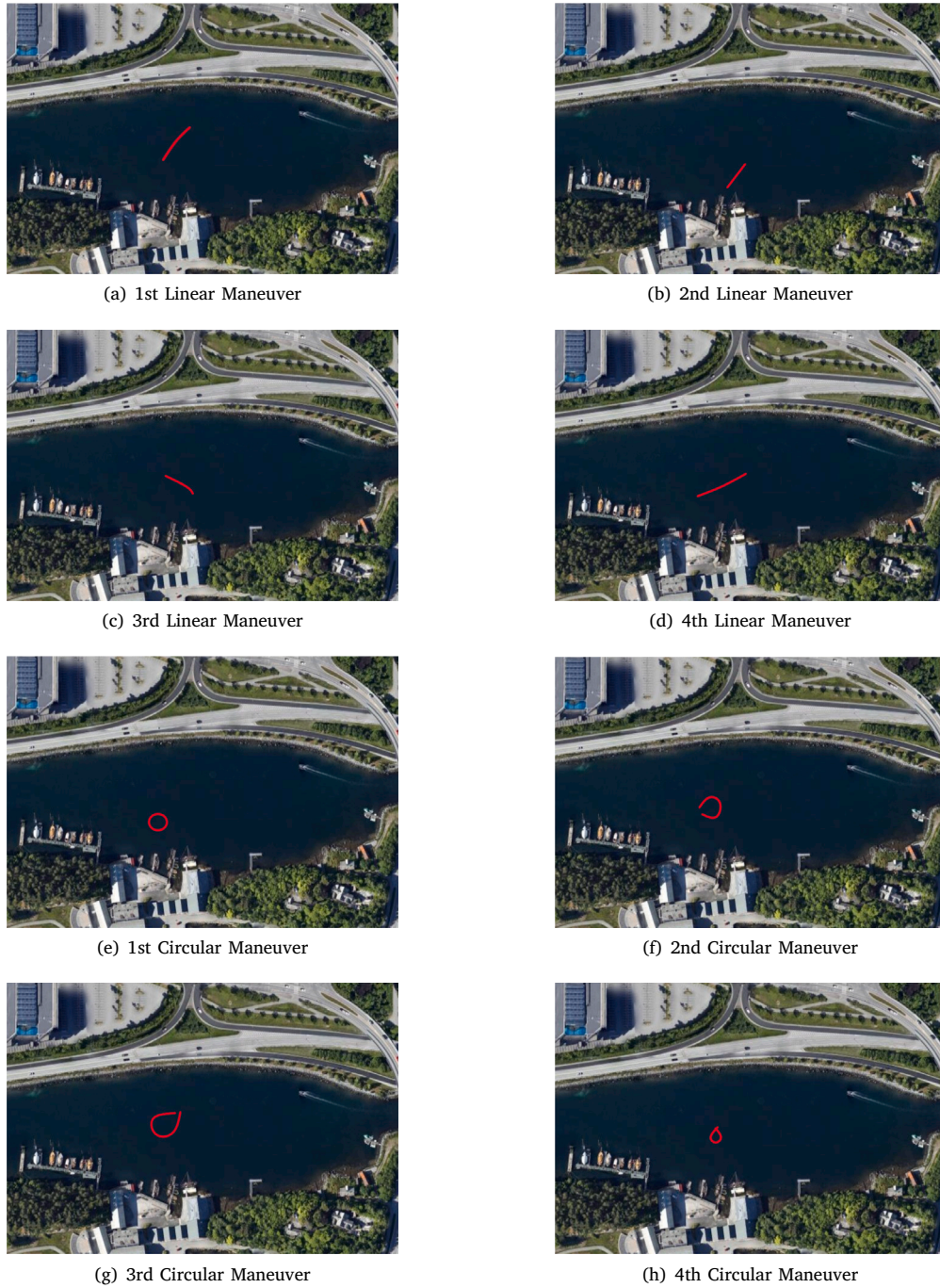


Fig. 1. The experiment comprised four linear and four circular maneuvers, designed to identify the parameters of the USV's dynamic model. These maneuvers were conducted under real-world conditions, and their trajectories are shown in red on a map of the test area.

Table 1

Library types and states.

Library Type	States
1 Degree Polynomial	$1, u, v, r, T_u, T_r$
2 Degree Polynomial	$1, u, v, r, T_u, T_r, u^2, uv, ur, uT_u, uT_r, v^2, vr, vT_u, vT_r, r^2, rT_u, rT_r, T_u^2, T_uT_r, T_r^2$
Custom Library	$u, v, r, T_u, T_r, ur, vr, r^2$

2.2. Experimental equipment

In this study, we utilize the Otter USV by Maritime Robotics. This USV, depicted in Fig. 2, is designed for various tasks in offshore operations such as research, hydrodynamic visualization, and environmental monitoring. It can be operated both autonomously and controlled remotely. The USV is equipped with two AC motors, producing up to 125 Newtons with high performance rechargeable lithium-ion batteries to power the motors, processors, and sensors. The USV has an advanced onboard computer to control and monitor the system. Onboard sensors include GNSS receiver and antenna, AIS receiver, VHF whip antenna,



Fig. 2. The Otter uncrewed surface vehicle (USV) employed in the system identification experiments is equipped with GPS and additional onboard sensors for recording position, velocity, acceleration, and heading data required for dynamic model parameter estimation.

4G dongle/modem and antenna, WiFi interface with 2.4 GHz antenna, and high definition camera.

Combining an inertial measurement unit (IMU) with Global Navigation Satellite System (GNSS) technology offers significant advantages over using GNSS alone. Integrating an IMU with GNSS improves positioning accuracy by compensating for temporary disruptions in GNSS signals. The IMU measures the device's acceleration and angular rate, allowing the system to provide continuous position and orientation estimates even in signal obstructions or interference. This combination of IMU and GNSS data enhances accuracy by filling in gaps during GNSS outages or degraded signal conditions. GNSS receivers typically offer position updates at relatively low rates, while IMUs can provide updates at much higher frequencies.

By combining these two units, the resulting system can provide more frequent and smoother position updates, which is vital for system identification. The combined systems offer faster initialization compared to GNSS only systems. GNSS receivers generally require some time to connect with satellites and determine the initial position. The system can significantly reduce initialization time with an IMU integration, which can provide a rapid estimate of the initial position and velocity. Although the initialization time does not take too much, it occupies space on the log files. Therefore, a more computational resource is required for pre-processing. The integration of IMU and GNSS measurements enables precise attitude determination, which is especially important for this dissertation due to the dynamic equation of the system. While GNSS alone provides position and velocity information, it lacks precise orientation (attitude) data when the system is stationary. By integrating IMU measurements, the system can accurately determine the device's yaw angles, which are essential for the experiment.

2.3. Dynamic modeling of the uncrewed surface vehicle

The equations of motion for a surface vehicle in three degrees of freedom can be described by the following model (Fossen, 2002):

$$\dot{\eta} = R(\psi)v \quad (1)$$

$$M\dot{v} = -(C(v) + D(v))v + B_r\tau_c + \tau_w \quad (2)$$

where $\eta = [x, y, \psi]^T$ is the vehicle pose in the North East Down (NED) frame. Here, x is the North, y is the East coordinate, and ψ is the heading angle as shown in Fig. 3(b). Similarly, $v = [u, v, r]^T$ is the

generalized velocity vector in the body frame, which are surge velocity, sway velocity, and yaw rate, respectively, as depicted in Fig. 3(a). In Eq. (1), $R(\psi)$ is the following rotation matrix, transforming vectors in the body-fixed frame to NED:

$$R(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

In Eq. (2), M is the mass matrix, which can be considered a superposition of two matrices, as seen in Eq. (4). The first is for the rigid body motion (M_{RB}), and the second matrix is the added mass matrix (M_A). Some surrounding water also has to accelerate with the vessel, resulting in hydrodynamic added mass. This effect is superposed over rigid body dynamics.

$$M = M_{RB} + M_A \quad (4)$$

M_{RB} and M_A are given by:

$$M_{RB} = \begin{bmatrix} m & 0 & -my_g \\ 0 & m & mx_g \\ -my_g & mx_g & I_z \end{bmatrix} \quad (5)$$

$$M_A = \begin{bmatrix} -X_{\dot{u}} & -X_{\dot{v}} & -X_{\dot{r}} \\ -Y_{\dot{u}} & -Y_{\dot{v}} & -Y_{\dot{r}} \\ -N_{\dot{u}} & -N_{\dot{v}} & -N_{\dot{r}} \end{bmatrix} \quad (6)$$

In Eq. (5), m represents the vehicle mass, and I_z is the moment of inertia around the z -axis in the body frame. x_g and y_g are the distances of the origin of the body axis to the center of gravity in x -body and y -body axis (Fossen, 2011b). In Eq. (6), SNAME (1950) (Society of Naval Architects and Marine Engineers (U.S.). Technical and Research Committee. Hydrodynamics Subcommittee, 1950) notation is used for hydrodynamic derivatives. For example, the first element in Eq. (6), represent the added mass force in X direction due to the acceleration in x direction, where the acceleration is denoted by $\dot{u}$ in x_b direction. Explicit notation of a derivative is given in Eq. (7) (Fossen, 2011a).

$$X_{\dot{u}} = \frac{\partial X}{\partial \dot{u}} \quad (7)$$

The mass matrix becomes:

$$M = \begin{bmatrix} m - X_{\dot{u}} & -X_{\dot{v}} & -my_g - X_{\dot{r}} \\ -Y_{\dot{u}} & m - Y_{\dot{v}} & mx_g - Y_{\dot{r}} \\ -my_g - N_{\dot{u}} & mx_g - N_{\dot{v}} & I_z - N_{\dot{r}} \end{bmatrix} \quad (8)$$

The C matrix in Eq. (2) represents the Coriolis and centrifugal matrix. Similar to the mass matrix, the Coriolis and centrifugal matrix is composed of two different matrices given in Eq. (9), which are $C_{RB}(v)$ and $C_A(v)$. The notation $C_{RB}(v)$ denotes the rigid body Coriolis and centrifugal matrix due to the rotation body axis about the inertial frame, while $C_A(v)$ is the hydrodynamic Coriolis and centrifugal matrix (Fossen, 2011a).

$$C(v) = C_{RB}(v) + C_A(v) \quad (9)$$

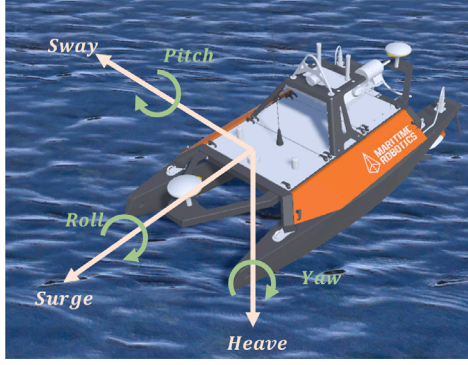
Here, C_{RB} and C_A are given by Fossen (2011b):

$$C_{RB}(v) = \begin{bmatrix} 0 & 0 & -m(x_g r + v) \\ 0 & 0 & mu \\ m(x_g r + v) & -mu & 0 \end{bmatrix} \quad (10)$$

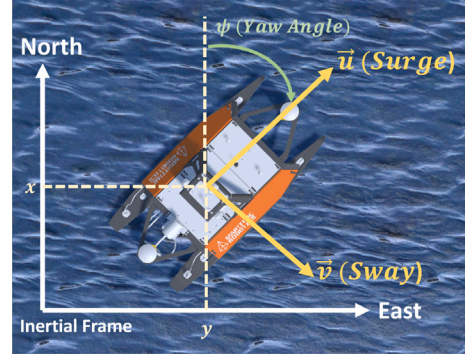
$$C_A(v) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u \\ -Y_{\dot{v}}v - Y_{\dot{r}}r & X_{\dot{u}}u & 0 \end{bmatrix} \quad (11)$$

The hydrodynamic derivative notation is explicitly shown in Eq. (7). Thus, the Coriolis centripetal matrix becomes:

$$C(v) = \begin{bmatrix} 0 & 0 & -m(x_g r + v) + Y_{\dot{v}}v + Y_{\dot{r}}r \\ 0 & 0 & mu - X_{\dot{u}}u \\ m(x_g r + v) - Y_{\dot{v}}v - Y_{\dot{r}}r & -mu + X_{\dot{u}}u & 0 \end{bmatrix} \quad (12)$$



(a) Six degrees of freedom (DOF) associated with the motion of the USV, illustrating translational motions (surge, sway, heave) and rotational motions (roll, pitch, yaw) relative to the body-fixed frame.



(b) Top-down view of the inertial (Earth-fixed) reference frame, showing the orientation of the north (Y) and east (X) axes. The USV's center of gravity (CG) is indicated, along with the surge, sway, and yaw directions relative to the body-fixed frame.

Fig. 3. Representation of the USV motion in both body-fixed and inertial frames. **Fig. 3(a)** illustrates the six degrees of freedom (DOF), including translational and rotational motions. **Fig. 3(b)** presents the inertial frame from a top-down view, with axes orientation and body-fixed motions shown at the center of gravity.

In Eq. (2), the $\mathbf{D}$ matrix represents the hydrodynamic damping effects. There are several reasons for the damping, such as potential damping, skin friction, wave drift damping, damping due to vortex shedding, etc. The damping matrix has both linear and nonlinear terms. Although it is challenging to separate those from each other, for many applications, the hydrodynamic damping matrix can be investigated in two parts: linear and nonlinear damping matrices (Fossen, 2011a).

$$\mathbf{D}(\mathbf{v}) = \mathbf{D} + \mathbf{D}_n(\mathbf{v}) \quad (13)$$

In Eq. (13), the $\mathbf{D}$ matrix represent the linear damping matrix due to potential damping and skin friction, and $\mathbf{D}_n(\mathbf{v})$ is the quadratic damping and the higher order terms (Fossen, 2011a). These matrices are given by Fossen (2011a):

$$\mathbf{D} = - \begin{bmatrix} X_u & 0 & 0 \\ 0 & Y_v & Y_r \\ 0 & N_v & N_r \end{bmatrix} \quad (14)$$

$$\mathbf{D}_n(\mathbf{v}) = - \begin{bmatrix} X_{|u|u}|u| & 0 & 0 \\ 0 & Y_{|v|v}|v| + Y_{|r|v}|r| & Y_{|v|r}|v| + Y_{|r|r}|r| \\ 0 & N_{|v|v}|v| + N_{|r|v}|r| & N_{|v|r}|v| + N_{|r|r}|r| \end{bmatrix} \quad (15)$$

In Eq. (15), similar SNAME (1950) (Society of Naval Architects and Marine Engineers (U.S.). Technical and Research Committee. Hydrodynamics Subcommittee, 1950) notation is used. The Damping matrix becomes:

$$\mathbf{D}(\mathbf{v}) = \begin{bmatrix} X_u + X_{|u|u}|u| & 0 & 0 \\ 0 & Y_v + Y_{|v|v}|v| + Y_{|r|v}|r| & Y_r + Y_{|v|r}|v| + Y_{|r|r}|r| \\ 0 & N_v + N_{|v|v}|v| + N_{|r|v}|r| & N_r + N_{|v|r}|v| + N_{|r|r}|r| \end{bmatrix} \quad (16)$$

For an underactuated system in 3DoF, there are two control inputs, $\tau_c = [T_u, \tau_r]^T$. T_u is the forward thrust and τ_r represent the torque applied on the vehicle.

2.4. Sparse identification of nonlinear dynamics (SINDy)

A data driven modeling method called SINDy was developed to discover the governing equations that underlie the dynamics of a complex system. The methodology employed in this study follows the approach outlined in Brunton et al. (2016), with necessary adaptations made in equations to represent the dynamics of USV. The same notation in mathematical equations is selected to achieve ease in comprehension, as discussed in Brunton et al. (2016). The foundation of SINDy is that

most dynamic systems can be modeled by a set of partial or ordinary differential equations. The nonlinear elements in these equations are crucial in defining how the system behaves. The nonlinear system can be represented as Eq. (17).

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)) \quad (17)$$

Here, $\mathbf{x}(t)$ represents the states at time t and $\mathbf{f}(\mathbf{x}(t))$ is the nonlinear function that includes the equation of motion. In this case:

$$\mathbf{x}(t) = [x_1(t), x_2(t), x_3(t), x_4(t), \dots, x_n(t)] \in \mathbb{R}^n \quad (18)$$

Sparsity promoting regression techniques are used to find the terms that best explain the observed dynamics to build a set of candidate functions that could appear in the governing equations. The sparsity constraint guarantees that the final model is as simple as possible while still accurately expressing the system's behavior. For the SINDy application, a set of time series data is required, which represents the system dynamics. The measured states Equation (19) and their derivatives Equation (20) are sampled in time instances (Brunton et al., 2016).

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}^T(t_1) \\ \mathbf{X}^T(t_2) \\ \vdots \\ \mathbf{X}^T(t_n) \end{bmatrix} = \begin{bmatrix} x_1(t_1) & x_2(t_1) & \dots & x_n(t_1) \\ x_1(t_2) & x_2(t_2) & \dots & x_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(t_m) & x_2(t_m) & \dots & x_n(t_m) \end{bmatrix} \quad (19)$$

$$\dot{\mathbf{X}} = \begin{bmatrix} \dot{\mathbf{X}}^T(t_1) \\ \dot{\mathbf{X}}^T(t_2) \\ \vdots \\ \dot{\mathbf{X}}^T(t_n) \end{bmatrix} = \begin{bmatrix} \dot{x}_1(t_1) & \dot{x}_2(t_1) & \dots & \dot{x}_n(t_1) \\ \dot{x}_1(t_2) & \dot{x}_2(t_2) & \dots & \dot{x}_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ \dot{x}_1(t_m) & \dot{x}_2(t_m) & \dots & \dot{x}_n(t_m) \end{bmatrix} \quad (20)$$

Then it is used to construct a library of candidate functions (Eq. (21)), including polynomial terms, trigonometric functions, or other nonlinear operators.

$$\Theta(\mathbf{X}) = [1 \quad \mathbf{X} \quad \mathbf{X}^{P_2} \quad \mathbf{X}^{P_3} \quad \dots \quad \cos(\mathbf{X}) \quad \sin(2\mathbf{X})] \quad (21)$$

where $\mathbf{X}^{P_k}$ denotes the higher order polynomials in the order of k .

The regression algorithm is then applied to this library to identify the sparsest set of terms that accurately capture the system's dynamics. In regression algorithm, there is a lot of flexibility in choosing the values for the matrix of nonlinearities. It is anticipated that only a few of these values is significant in each row of $\mathbf{f}(\mathbf{x}(t))$. Therefore, sparse regression can be employed to determine which values are significant by identifying the sparse vectors of coefficients $\Xi = [\xi_1 \quad \xi_2 \quad \dots \quad \xi_n]$.

$$\dot{X} = \Theta(X)\Xi \quad (22)$$

In Eq. (22), $\dot{X}$ is the derivative of the state vector, Θ is the candidate libraries, and Ξ is the coefficients of the sparse vectors (Brunton et al., 2016). In order to solve Eq. (22), the following optimization function is used to minimize (Kaiser et al., 2018):

$$\min \|\dot{X}_k - \xi_k \Theta^T(X)\|_2 + \lambda \|\xi\|_1 \quad (23)$$

Several ways to solve the above optimization problem exist, such as LASSO (Tibshirani, 1996) or sequentially thresholded least squares (Brunton et al., 2016).

2.5. Model validation and performance metrics

Validation in a machine learning model is highly significant in assessing its performance and generalization ability. The primary goal is to monitor how well the model will work with new, unseen data and determine if it meets the requirement for the deployment. Through model validation, valuable insights are obtained regarding its effectiveness and identification of potential challenges like overfitting or underfitting. Validation ensures the optimum model with fine tuned parameters. There are several commonly used techniques for model validation:

- **Holdout Validation (Train-Validation-Test Split):** The dataset is divided into three subsets: train, validation, and test sets. The model is trained using the training set and evaluated on the validation set. This approach provides a quick assessment of the model's performance. However, it is essential to note that holdout validation can be sensitive to the specific data split, mainly when dealing with time series data where random data splitting is not suitable (Shalev-Shwartz and Ben-David, 2014).
- **K-fold Cross Validation:** The dataset is divided into multiple folds or subsets. The model is trained on a combination of these folds and evaluated on the remaining fold. This process is repeated numerous times, with each fold serving as the validation set once. Cross validation provides a more robust estimate of the model's performance using different data subsets for validation (Nilsson, 2005).

Different evaluation metrics are typically employed to assess the model's performance. These metrics are designed explicitly for evaluating regression models, and their interpretation differs from those used in problems. Selecting the relevant statistical measures that align with the specific requirements is crucial when working on a regression problem. For this study Root Mean Square Error (RMSE) is selected. The RMSE provides an interpretable measure of the average prediction error in the same scale of the target variable (Botchkarev, 2019).

3. Results and discussion

In selecting a methodology for discovering the dynamical equations of the USV, several approaches can be considered, each with its own benefits and drawbacks. Below, discussion of alternative system identification and model discovery techniques are given, followed by a comparison showing why SINDy is particularly suitable for this application.

An application of a deep learning approach to modeling USV dynamics was presented in the study (Woo et al., 2018), where recurrent LSTM networks were applied to capture the vessel's motion. Their results demonstrated substantial reductions in prediction error for surge speed, yaw rate, and sway velocity compared with conventional linear models, highlighting the potential of such methods to capture strongly nonlinear, time-dependent behaviors. The advantage of these network-based models lies in their flexibility: they can represent complex couplings, actuator nonlinearities, and even environmental disturbances if

trained on sufficiently diverse datasets. However, their disadvantages are equally well recognized: the models are essentially black boxes, with limited interpretability, and they typically require large amounts of training data spanning the full operational envelope to generalize well. For a USV operating in varied conditions, this data requirement is often impractical, and the lack of transparency in the learned equations limits their usefulness for analysis or control system design. Another recent work (Dong et al., 2024) in marine system identification applies a time-varying parametric maneuvering model, estimating parameters in real time via a prediction error method with a dynamic window. This method adapts well to changes and provides real-time estimation and prediction. However, its performance depends heavily on how well the assumed parametric form captures the true dynamics. If some nonlinear terms are missing or not anticipated, those effects cannot be identified. Moreover, such methods may still be sensitive to disturbances, measurement noise, or unmodeled external forces, particularly if sensor fidelity is limited. Physics-informed neural networks (PINNs) represent another popular direction applied in several fields. In the context of maritime vehicles, the study (Xu et al., 2022) presents a PINN to identify surge, sway, and yaw dynamics of a USV using zigzag and straight-line maneuvers. The authors show that, with limited training data, embedding physical constraints in the loss function yields better generalization than standard neural networks. Similarly, in Guan et al. (2024), PINNs are used to predict yaw motion of a vehicle with the aid of CFD data. These approaches offer a promising compromise between interpretability and flexibility, but even here the challenge remains in selecting which physical constraints or prior knowledge to include, and in handling noise in derivative estimates.

SINDy provides a balanced set of advantages that align well with the objectives of this study. Its primary strength lies in generating explicit, sparse equations that reflect physical dynamics (e.g., inertia, drag, coupling), making it more interpretable than methods such as deep neural networks or PINNs, which may obscure dominant terms. This interpretability aids not only in understanding the dynamics but also in control design, stability analysis, and vehicle modifications. SINDy also performs well with modest datasets, since its sparse regression encourages simpler models, unlike LSTMs or full neural networks, which require large training sets to avoid overfitting. This is particularly beneficial in marine field tests, where data collection is expensive and constrained by environmental conditions. In addition, SINDy allows the incorporation of prior knowledge through a candidate function library, offering flexibility without rigid parametric constraints. This reduces the risk of overfitting compared to black-box models, which may incorporate irrelevant or spurious terms. Nonetheless, SINDy is not without limitations. If the candidate library omits key dynamics such as nonlinear couplings, time delays, or hysteresis, these effects cannot be discovered. Furthermore, numerical estimation of derivatives can be sensitive to noise, and a fixed SINDy model may degrade under changing conditions unless adapted.

Considering the advantages and limitations of SINDy, as mentioned in the previous section, there are eight maneuvers specifically designed to excite the system and capture its complete dynamics. The maneuvers are classified into two categories, those performed along a linear path and those on a circular path. However, especially in the linear paths, additional excitement to the system is given to demonstrate the coupling effects in the motion. In the maneuvers Fig. 4, 5 and 6, an excitement is applied with different frequencies. Notably, the frequency in Fig. 4 is lower compared to Fig. 6. However, in Fig. 4 maneuver, a continuous thrust force is applied as input differing from Fig. 6. The Fig. 6 aims to capture the natural response of the system. On the other hand, in Fig. 7, a step input is introduced to the system instead of a periodic input. It should be noted that the maneuvers are given manually due to the system restriction. Therefore, some small imperfections may occur. In the figures, each maneuver is trained with three different library functions, which are given in the preceding section. Comparing the

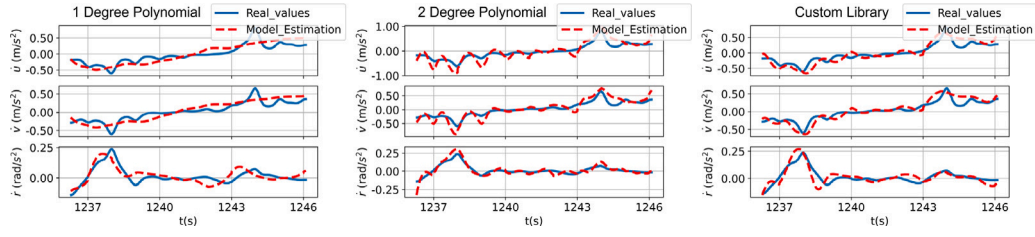


Fig. 4. Comparison of the SINDy model and experimental data for the first linear maneuver.

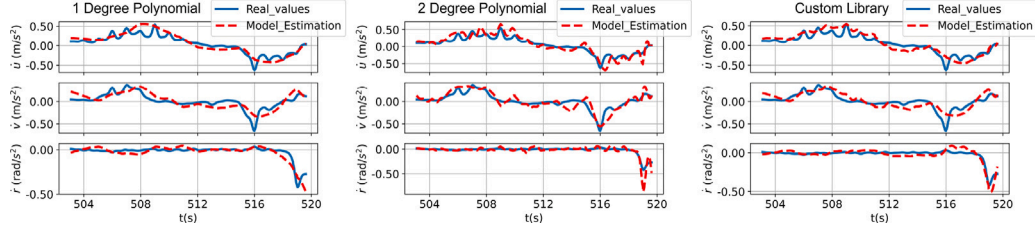


Fig. 5. Comparison of the SINDy model and experimental data for the second linear maneuver.

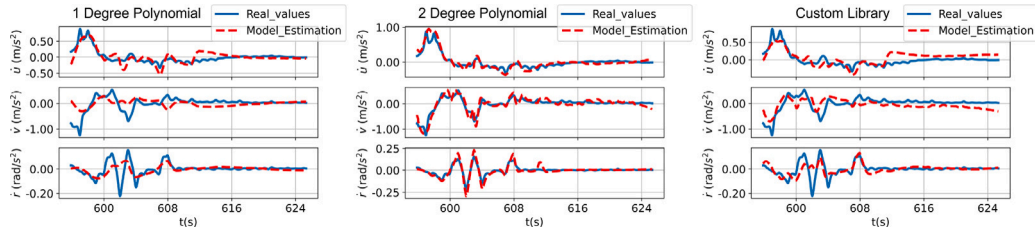


Fig. 6. Comparison of the SINDy model and experimental data for the third linear maneuver.

capability of each library 1st order polynomial library can capture the dynamics sufficiently if the inputs are smooth and the environment is undisturbed.

In Fig. 4, this can easily be seen that with low frequency input, the 1st order polynomial library effectively captures the dynamics. This can be supported with Table 5, as compared to the RMSE values of first linear maneuver in first and second order; even if the model complexity is increased significantly, the error metrics are almost the same supporting the argument. A more sophisticated model is required to capture the dynamics with rapid input changes. As depicted in Fig. 6, the model is perturbed with some rotational force. When the model is trained with a second order polynomial library, highly accurate estimations are obtained compared to first order polynomials. The effect is visible in $\dot{v}$ and $\dot{r}$ curves. Also, in Tables 3 and 4, corresponding RMSE values are almost reduced to half, which is a substantial improvement. Comparing the performance of models trained with linear maneuvers and circular maneuvers, it is evident that the models trained with circular maneuvers have higher accuracy, as demonstrated in Table 5. There are two reasons for this improvement. Firstly, the inputs were continuous during the circular maneuvers, resulting in smooth transitions between the states. Secondly, the fidelity of the lateral dynamics is higher in circular maneuvers since more data is provided for this dynamic. The coupling in the dynamic equation is more evident, allowing to capture of underlying relations with higher accuracy.

In the context of the circular maneuvers, four different circular maneuvers are conducted in the experiment as shown in Figs. 8, 9, 10, and 11. As mentioned in the preceding section, these maneuvers were in different velocities and radii. The circular maneuvers are more intuitive since they can simultaneously excite both longitudinal and lateral dynamics. On the other hand, they are more complex to interpret the complete dynamics. The RMSE values for $\dot{u}$ and $\dot{v}$ are not significantly altered by the library choice, in contrast to the linear

maneuver. The maximum disparity is only between 10 to 15 percent. Second order terms are introduced in the custom library mainly due to the presence of the Coriolis effect. It can be easily deduced that in such a small and slow system, the coefficients associated with the linear terms, such as mass, inertia, and hydrodynamic coefficients, are more dominant than those related to the Coriolis effect. Therefore, the impact of the quadratic terms on the overall dynamics is less significant. In Table 2, all the circular maneuvers have similar error metrics except fourth circular maneuver. This particular maneuver involves a tighter trajectory, i.e., conducted with the smallest radius among the rest. The centripetal force acting on the vehicle is much more dominant than tangential. Due to the dominance of centripetal force, the acceleration along the body's x-axis experiences less variation than the acceleration along the y-axis. Therefore, the RMSE value for acceleration in the body x-axis is lower than the y-axis. First circular maneuver is intended to have the same normal acceleration as fourth circular maneuver with higher tangential acceleration which can be observed in Figs. 8 and 11. Although the RMSE values in Table 3 show similar results, there is a significant difference in the error metric in Table 2. The error metric for first circular maneuver is approximately 40 percent higher than for fourth circular maneuver. This finding suggests that not only the shape but also the magnitude of the input signal impacts the model discovery. In contrast to the linear maneuvers, where additional excitations are given to the system, the circular maneuvers maintain a constant input to guide the vehicle along a circular trajectory. Due to the continuous input, a smoother response from the system is obtained. This can be observed from all four circular maneuver curves by their smooth wave function shape, particularly in Figs. 9 and 10. When comparing the RMSE results in Table 5, the circular maneuvers tend to have less error metric than linear maneuvers. This observation shows that continuous and smooth inputs produce a more accurate model when using the SINDy approach for model identification.

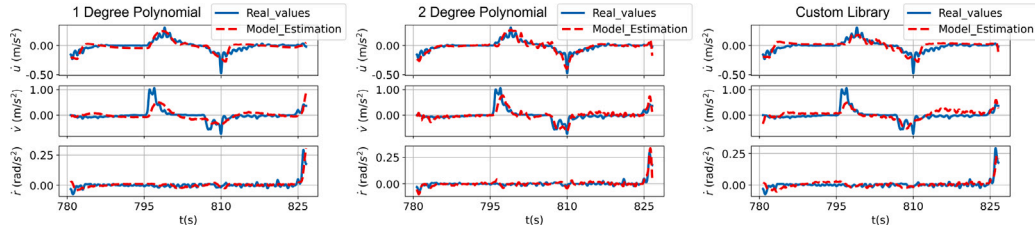


Fig. 7. Comparison of the SINDy model and experimental data for the fourth linear maneuver.

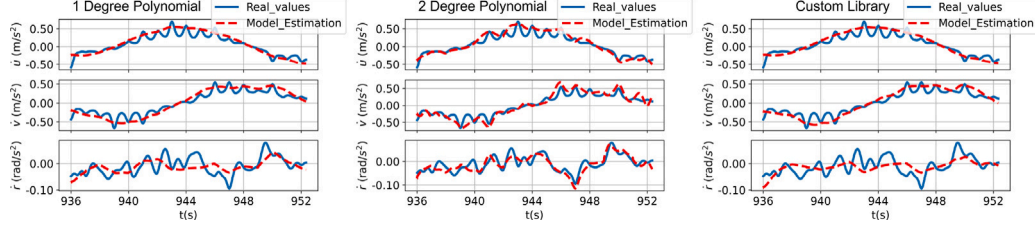


Fig. 8. Comparison of the SINDy model and experimental data for the first circular maneuver.

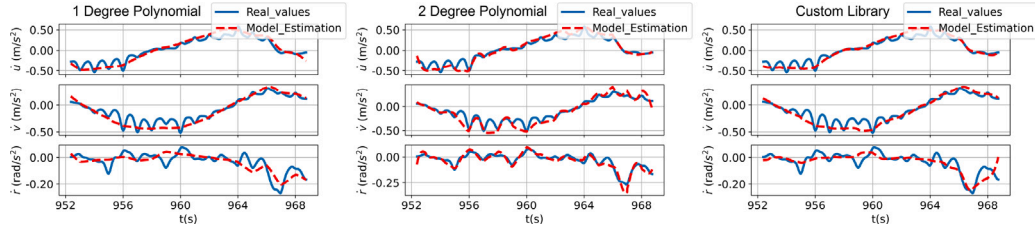


Fig. 9. Comparison of the SINDy model and experimental data for the second circular maneuver.

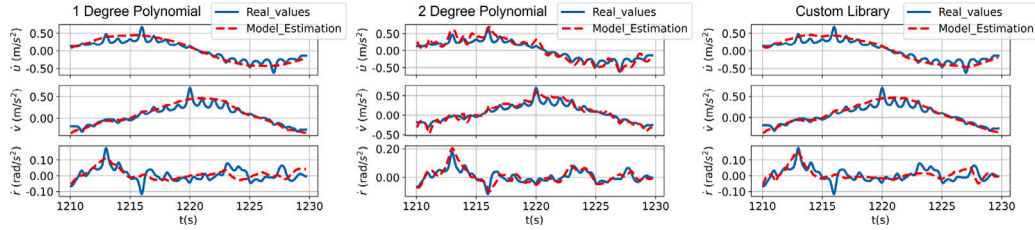


Fig. 10. Comparison of the SINDy model and experimental data for the third circular maneuver.

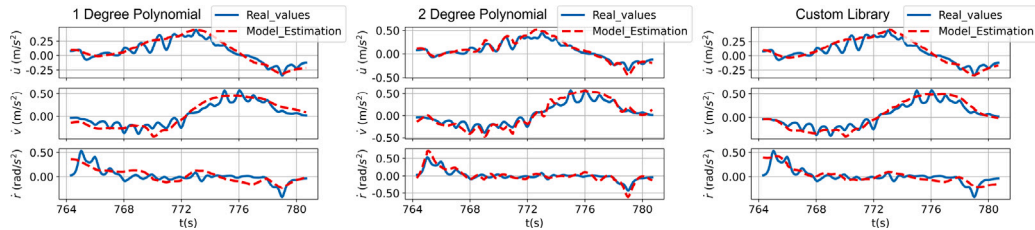


Fig. 11. Comparison of the SINDy model and experimental data for the fourth circular maneuver.

Two different validation method is used in the study. In the first method, a modal is trained with one maneuver and tested with the other remaining maneuvers. This validation method mainly aims to identify the maneuvers accurately representing the complete dynamics. The second method used the k-fold validation, where each maneuver is treated as a fold, and one of the maneuvers is kept as test data in each run. This method is used to have a comprehensive understanding of the data collected. When the validation Tables 6–8 are observed, it becomes apparent that although the second order polynomial library

performs well in the training models, in the validation phase, they perform poorly. This indicates that the models trained with second order polynomial libraries are prone to over-fitting. This library is intended to be the black box model where no prior knowledge exists about the dynamics and aiming to discover new parameters that did not exist in literature. However, the average error metrics in Table 7 are considerably larger, associated with high standard deviation, suggesting a lack of generalizability in these models. Especially when the model is

Table 2
RMSE value of $\dot{u}$ with training dataset.

Maneuvers	1st Order	2nd Order	Custom Library
Circular 1	0.109	0.105	0.109
Circular 2	0.090	0.090	0.088
Circular 3	0.099	0.110	0.099
Circular 4	0.069	0.066	0.070
Linear 1	0.154	0.164	0.132
Linear 2	0.109	0.118	0.110
Linear 3	0.129	0.079	0.142
Linear 4	0.053	0.043	0.059

Table 3
RMSE value of $\dot{v}$ with Training Dataset.

Maneuvers	1st Order	2nd Order	Custom Library
Circular 1	0.113	0.118	0.113
Circular 2	0.086	0.093	0.086
Circular 3	0.078	0.081	0.078
Circular 4	0.105	0.091	0.095
Linear 1	0.131	0.137	0.113
Linear 2	0.108	0.092	0.106
Linear 3	0.263	0.121	0.231
Linear 4	0.170	0.143	0.166

trained with circular maneuvers and tested with the remaining maneuvers, the RMSE values are too high compared to the models trained with linear maneuvers. This demonstrates that the more complex dynamics suffer more severely from over-fitting than the simpler motions. When [Tables 6 and 8](#) are compared, it can be concluded that the first order models trained with linear maneuvers are more generalizable than the custom library. The average RMSE value in first order linear maneuvers models reaches the maximum of 1.200 with a standard deviation of 0.704. In contrast, the custom library has a minimum average of 4.272 with a standard deviation of 4.437. However, upon closer inspection of [Table 8](#), the high values are introduced due to the validation with circular maneuvers. This is expected since the lateral dynamics are not adequately excited during the training. Therefore, claiming the first order polynomial library is a better option would be incorrect.

The models trained with circular maneuvers perform better with the custom library compared to the first order library. The average error metric does not exceed 1, with a maximum standard deviation of 0.691. The circular maneuver contains more complex dynamics, including longitudinal and lateral motion. The model trained with a custom library performs better both in training and validation if the maneuvers are wisely chosen to excite the complete dynamics. In [Table 9](#), the k-fold validation results for the maneuvers are presented. The first notable observation is on the linear maneuver 1 for all the function libraries; the RMSE value is significantly higher than the mean of the error metrics. This increase in error metrics also yields an increase in standard deviation. This indicates that during this maneuver, some unexpected disturbances occurred during the experiment. However, a detailed investigation on these disturbances cannot be conducted due to a lack of physical sensors. A reasonable interpretation of this outcome is the vehicles passing by the USV induce additional disturbances, such as waves. Other factors, such as the flow of the sea or wind, do not change in a short amount of time. This argument is also reinforced by [Table 8](#).

The model trained with the first linear maneuver has the highest error metric when validated with other linear maneuvers. It should be noted that the validation of circle maneuvers using the model trained in linear maneuvers does not yield meaningful results, which is explained in detail previously. Another finding that would be obtained from [Table 9](#) is the performance of the custom library compared to the other libraries. As mentioned before, the second order library is over-fitting and lacks generalizability. The average error metric and standard

Table 4
RMSE value of $\dot{r}$ with training dataset.

Maneuvers	1st Order	2nd Order	Custom Library
Circular 1	0.027	0.018	0.028
Circular 2	0.046	0.026	0.040
Circular 3	0.035	0.028	0.034
Circular 4	0.108	0.081	0.095
Linear 1	0.043	0.033	0.042
Linear 2	0.039	0.055	0.038
Linear 3	0.045	0.024	0.043
Linear 4	0.023	0.017	0.022

Table 5
RMSE value of the model with training dataset.

Maneuvers	1st Order	2nd Order	Custom Library
Circular 1	0.083	0.080	0.083
Circular 2	0.074	0.070	0.071
Circular 3	0.071	0.073	0.070
Circular 4	0.094	0.079	0.087
Linear 1	0.109	0.111	0.096
Linear 2	0.085	0.088	0.085
Linear 3	0.146	0.075	0.138
Linear 4	0.082	0.068	0.082

deviation in the second order library are the highest. On the other hand, the first order library performs relatively better and has the lowest standard deviation. A lower standard deviation in the RMSE values is desirable; however, the metric average is 26.8 percent higher than a custom library. Despite a slight increase in standard deviation in the custom library, the error metric significantly reduces. Also, with the elimination of the outlier maneuver, the standard deviation can be reduced to 0.051. Consequently, it can be deduced that the grey model approach provides a more robust and generalizable model, where some prior knowledge is inherited through a custom library function.

Unlike the other methods in the literature, SINDy allows fusing several maneuvers into a single training set even if they are not conducted consequently. As a result, it enables robust model estimation even with streaming or partial data. In [Fig. 12](#), eight maneuvers are used to train the model, an additional maneuver, which is not introduced during model training, was used for the test case. The model receives three measurements from the physical asset: u , v , and r . [Fig. 12](#) demonstrates different update frequencies. Overall, the 6.25 Hz, 2.5 Hz, and 1.22 Hz frequencies yield promising results. However, as the data transfer frequency decreases, the error accumulation increases. This implies that although the model can accurately estimate the state derivatives for a given time, which is discussed in the previous paragraphs, some corrections are necessary with the sensor measurements to prevent integration errors in the model.

[Fig. 12](#) shows that the model estimation of the yaw rate is more precise compared to the translation velocities. The yaw rate estimation from the trained model is more reliable in longer intervals. On the other hand, it cannot be inferred that the equations for translation velocities derived from the SINDy model are insufficient. The primary reason is the error in these states accumulates much faster than the yaw rate. This disparity can be easily observed in [Fig. 12](#) with 0.30 Hz and 0.24 Hz. After each initialization (or correction), the model can estimate the states precisely, but after a few iterations error significantly increases. Additionally, by comparing the shape of the curves, it is evident that the model's behavior is similar to the experimental data.

The integration of sensor characteristics and environmental disturbances into the digital twin platform was considered essential for assessing the practical relevance of the present work. While the study

Table 6
1st order library holdout validation.

Train/Test	C1	C2	C3	C4	L1	L2	L3	L4	Mean	σ
C1	nan	0.146	0.470	1.249	1.395	1.325	1.970	1.741	1.185	0.608
C2	0.116	nan	3.209	5.056	4.449	6.009	7.276	7.781	4.826	2.456
C3	0.080	0.174	nan	0.335	0.694	0.853	0.900	0.833	0.541	0.339
C4	0.188	0.235	1.001	nan	1.479	0.495	1.494	0.320	0.718	0.563
L1	1.813	2.014	1.825	1.415	nan	0.367	1.623	1.468	1.245	0.704
L2	1.267	1.473	1.408	1.019	0.865	nan	1.352	1.287	1.058	0.477
L3	0.984	0.899	0.889	0.969	0.639	0.558	nan	0.545	0.643	0.308
L4	1.066	1.115	1.068	0.894	0.705	0.678	0.843	nan	0.758	0.345

Table 7
2nd order library holdout validation.

Train/Test	C1	C2	C3	C4	L1	L2	L3	L4	Mean	σ
C1	nan	1.786	295.397	880.784	795.555	930.507	1339.151	1392.414	805.085	471.594
C2	0.618	nan	1135.584	2514.896	2062.953	2548.358	3599.736	3833.256	2242.112	1243.649
C3	0.593	1.334	nan	220.682	243.816	192.779	333.104	310.323	186.005	125.576
C4	1.760	2.396	110.835	nan	72.121	8.232	52.840	4.071	35.785	40.363
L1	112.687	130.804	121.205	85.444	nan	34.370	217.150	89.449	96.917	65.123
L2	88.898	93.211	96.546	75.492	32.020	nan	58.165	16.318	53.107	35.133
L3	16.106	15.938	12.777	14.963	5.316	2.400	nan	3.787	7.883	6.030
L4	18.924	15.935	19.717	14.256	35.178	2.568	16.727	nan	14.912	10.779

Table 8
Custom library holdout validation.

Train/Test	C1	C2	C3	C4	L1	L2	L3	L4	Mean	σ
C1	nan	0.378	0.493	0.833	1.830	0.932	0.726	1.289	0.926	0.461
C2	0.110	nan	0.489	0.280	2.267	1.225	0.694	0.816	0.824	0.691
C3	0.098	0.294	nan	0.672	1.334	0.918	0.677	0.753	0.664	0.396
C4	0.170	0.191	0.790	nan	1.231	0.471	0.911	0.187	0.540	0.418
L1	24.054	17.453	23.040	23.304	nan	5.782	8.828	4.680	11.870	8.665
L2	13.863	14.426	14.930	11.292	3.617	nan	3.233	1.090	6.941	5.928
L3	10.514	9.310	8.097	10.587	0.830	0.609	nan	0.473	4.272	4.437
L4	10.967	7.881	9.391	10.481	2.381	0.752	1.656	nan	4.649	4.103

Table 9
k-fold validation of maneuvers with various libraries.

Library	C1	C2	C3	C4	L1	L2	L3	L4	Mean	σ
1st Degree	0.157	0.141	0.143	0.156	0.372	0.209	0.233	0.216	0.203	0.072
2nd Degree	0.090	0.103	0.177	0.385	0.646	0.194	0.333	0.216	0.268	0.172
Custom Library	0.081	0.088	0.098	0.118	0.352	0.175	0.235	0.135	0.160	0.087

primarily focused on demonstrating the feasibility of sparse identification for USV dynamics, the effects of measurement errors and external forces on the identified models cannot be overlooked when progressing toward the development of a digital twin. In this context, the digital environment can be extended to generate sensor outputs that reflect realistic conditions and to embed environmental disturbances that capture common operational challenges.

In the case of sensors, measurements may be represented as noisy and biased versions of the true states. Additive Gaussian noise has traditionally been used to approximate sensor inaccuracies, but more realistic models include low-frequency bias drift, correlated noise processes, and intermittent measurement loss. Such formulations enable the reproduction of phenomena such as GPS signal dropouts, IMU drift, and compass disturbances within the digital twin environment. By subjecting the identification framework to these imperfections, the robustness of the recovered models can be evaluated under conditions closer to practical deployment. Furthermore, latency and synchronization mismatches between different sensing modalities may be introduced to assess their influence on the consistency of the identified dynamics (Alqurashi et al., 2023). Environmental effects can likewise be incorporated into the governing equations as external disturbances. Wind forcing may be represented using aerodynamic drag models, with stochastic variations added to capture gusts. Surface currents may be introduced as low-frequency biases in surge and sway velocities, or as

time-varying disturbances described by random-walk or spectral models. Wave loads may be represented either through simplified sinusoidal forcing at selected frequencies or through stochastic wave spectra that generate oscillatory disturbances in both translational and rotational degrees of freedom. When included in the digital twin platform, these disturbances act as unmeasured external inputs and directly test the capability of the identification algorithm to distinguish intrinsic vessel dynamics from externally driven behavior (Gu et al., 2022).

Simulation frameworks such as the Virtual RobotX (VRX) platform have been established as benchmarks for USV research. For example, in Jin et al. (2025), a simulation system was developed by combining VRX and Simulink/ROS2 for a WAM-V USV with podded propellers, with wind effects incorporated and controllers for yaw rate, heading, and path-following modes tested. Water tank experiments were also conducted to validate performance. A related line of research is presented in Yoon and Kim (2024), where parameter identification for kinetic models of a dual-thruster USV in the VRX simulation environment was carried out, and close alignment between the kinetic model and VRX trajectories was achieved. The present work is positioned as a complement to these efforts by enabling a data-driven discovery of USV dynamics through SINDy, applied directly to input/output data, rather than relying solely on pre-specified hydrodynamic coefficients or rigid simulator models. Although VRX-based simulations are widely recognized as powerful tools for input testing, navigation, perception, and

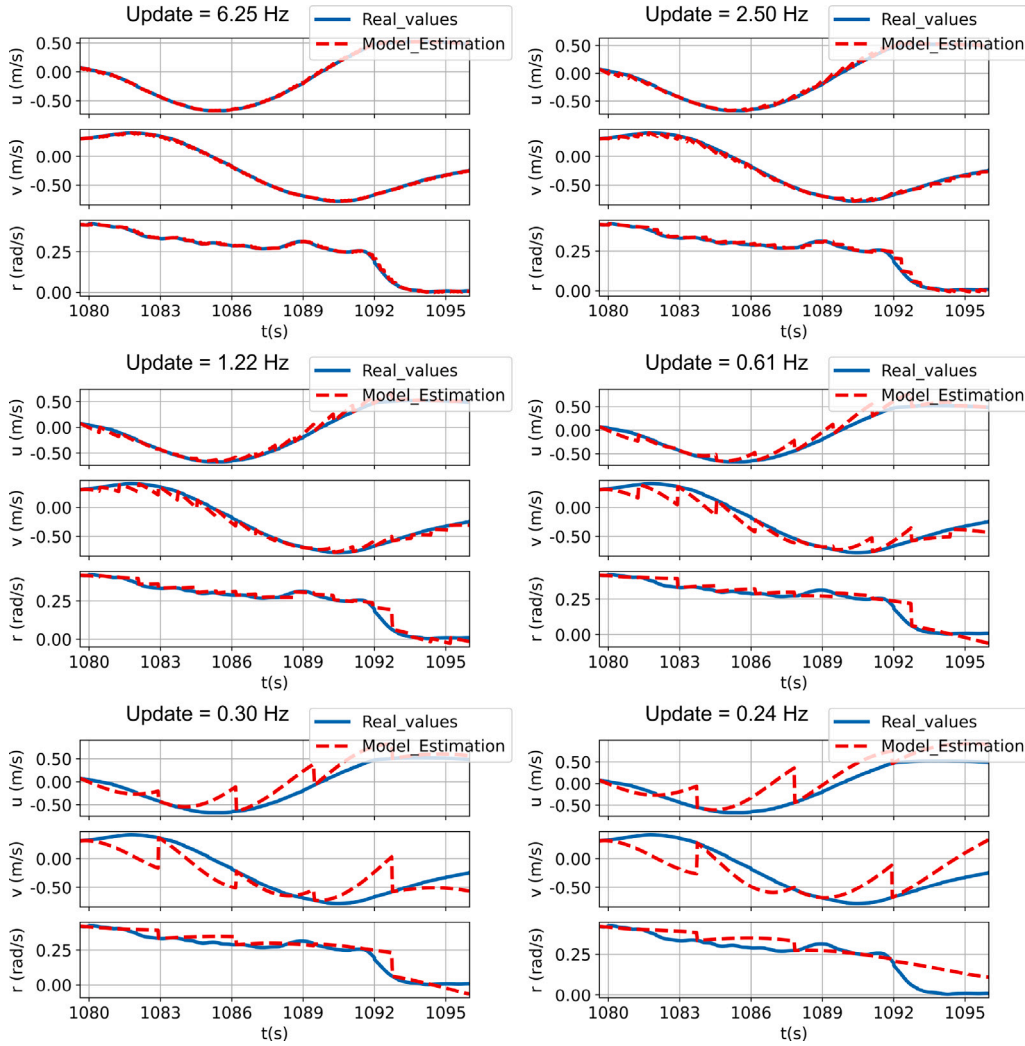


Fig. 12. SINDy model trained with eight maneuvers and tested on an independent maneuver. Model predictions for u , v , and r are shown under different update frequencies. Higher update rates (6.25–1.22 Hz) yield accurate estimates, while lower rates (0.30–0.24 Hz) cause error accumulation. Yaw rate predictions remain more reliable than translational velocities, though overall trends follow the experimental data.

control algorithm benchmarking, their reliance on particular parameterizations may result in the omission of platform-specific nonlinearities that only emerge from real or high-fidelity data. Such nonlinearities can be identified by SINDy, which offers parsimonious models that may adapt more effectively to diverse operating conditions.

4. Conclusions and further development

This study applies the Sparse Identification of Nonlinear Dynamical Systems (SINDy) framework to derive the governing dynamics of a physics-based digital twin for an uncrewed surface vehicle (USV). Model accuracy is evaluated through cross-validation at both the individual state level and for the complete system. To enhance robustness and interpretability, prior knowledge of the USV's dynamics is incorporated into the model identification process, resulting in improved error metrics. This integration provides deeper insight into the vehicle's complex behavior while enhancing model transparency. However, while the flexibility of the chosen function libraries supports model generalizability, it may also limit physical interpretability when dynamics fall outside predefined terms. In addition, because the model relies on numerical differentiation of variables, measurement noise can lead to inaccuracies.

The practical implications are significant for ocean practice. Em-

bedding such models into physics-based digital twins can support safer and more reliable autonomous navigation, predictive maintenance, and mission planning in offshore operations. This capability is especially relevant for long-duration environmental monitoring campaigns, inspection of offshore infrastructure, and operations in hazardous or remote marine environments where real-time decision support is essential.

Future research should extend this work in several directions. First, the inclusion of additional sensors to capture environmental effects such as water currents, wind direction, and velocity could further refine model accuracy. The inclusion of such sensor and environmental models is not only expected to improve the realism of the digital representation but also provides a pathway for evaluating the robustness of sparse regression methods such as SINDy. The amplification of noise during numerical differentiation, the risk of false term selection under bias drift, and the possibility of missed dynamics due to oscillatory disturbances represent challenges that can be systematically examined in this extended framework. Second, employing advanced technologies such as Real-Time Kinematic (RTK) GPS and higher-frequency sensors would improve the precision and reliability of collected data, thereby supporting the evaluation of model effectiveness for autonomous guidance and control. Thirdly, high-fidelity simulators such as VRX may be employed to generate diverse trajectories. SINDy can then be used to

identify correction or residual models on top of baseline physics-based USV models. The resulting models can be validated in both simulation and real sea trials to assess their fidelity and generalization. Finally, scaling digital twin frameworks from individual vessels to coordinated fleets represents a critical step toward enabling collaborative ocean observation and multi-vehicle mission planning.

CRedit authorship contribution statement

Ege Kandemir: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Erlend Magnus Lervik Coates:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Agus Hasan:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Alexandersson, M., Mao, W., Ringsberg, J.W., 2022. System identification of vessel manoeuvring models. *Ocean Eng.* 266, 112940.
- Alqurashi, F.S., Trichili, A., Saeed, N., Ooi, B.S., Alouini, M.-S., 2023. Maritime communications: A survey on enabling technologies, opportunities, and challenges. *IEEE Internet Things J.* 10 (4), 3525–3547.
- Botchkarev, A., 2019. A new typology design of performance metrics to measure errors in machine learning regression algorithms. *Interdiscip. J. Inf. Knowl. Manag.* 14, 45–79.
- Brunton, S.L., Proctor, J.L., Kutz, J.N., 2016. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proc. Natl. Acad. Sci.* 113 (15).
- Dong, Z., Wang, B., Tan, F., Zhou, W., Liu, Y., 2024. Online parameter identification and real-time manoeuvring prediction for a water-jet USV based on weighted multi-innovation prediction error method integrated with dynamic window strategy. *Appl. Ocean Res.* 153, 104260.
- Fossen, T.I., 2002. *Marine Control Systems: Guidance, Navigation and Control of Ships, Rigs and Underwater Vehicles*. Marine Cybernetics, Trondheim, Norway.
- Fossen, T.I., 2011a. Manoeuvring theory. In: *Handbook of Marine Craft Hydrodynamics and Motion Control*. John Wiley & Sons, Ltd, UK, pp. 109–132.
- Fossen, T.I., 2011b. Models for ships, offshore structures and underwater vehicles. In: *Handbook of Marine Craft Hydrodynamics and Motion Control*. John Wiley & Sons, Ltd, UK, pp. 133–186.
- Gu, N., Wang, D., Peng, Z., Wang, J., Han, Q.-L., 2022. Disturbance observers and extended state observers for marine vehicles: A survey. *Control Eng. Pract.* 123, 105158.
- Guan, Y., Bian, Y., Zheng, H., Wang, X., Cui, Q., Ding, R., 2024. Prediction modeling for yaw motion of deep-sea mining vehicle during deployment and recovery: A physics informed neural network (PINN) approach. *Appl. Ocean Res.* 153, 104208.
- He, H.-W., Wang, Z.-H., Zou, Z.-J., Liu, Y., 2022. Nonparametric modeling of ship manoeuvring motion based on self-designed fully connected neural network. *Ocean Eng.* 251, 111113.
- Jin, P., Li, W., Yang, Y., Shan, C., Zhang, Y., 2025. Construction of simulation system for USV motion control and design of multi-mode controllers based on VRX and simulink. *Appl. Sci.* 15 (8).
- Kaiser, E., Kutz, J.N., Brunton, S.L., 2018. Sparse identification of nonlinear dynamics for model predictive control in the low-data limit. *Proc. R. Soc. A: Math. Phys. Eng. Sci.* 474 (2219), 20180335.
- Kandemir, E., 2023. *Sparse Identification of Nonlinear Dynamics in Digital Twins for an Unmanned Surface Vessel*. Norwegian University of Science and Technology.
- Liu, A., Xue, Y., Qin, H., Zhu, Z., 2024. Physics-informed identification of marine vehicle dynamics using hydrodynamic dictionary library-inspired adaptive regression. *Ocean Eng.* 296, 117013.
- Ljung, L., 2012. *System Identification: Theory for the User*. Prentice Hall PTR, Upper Saddle River, NJ.
- Luo, W., Guedes Soares, C., Zou, Z., 2016. Parameter identification of ship maneuvering model based on support vector machines and particle swarm optimization. *J. Offshore Mech. Arct. Eng.* 138 (3).
- Luo, W.L., Zou, Z.J., 2009. Parametric identification of ship maneuvering models by using support vector machines. *J. Ship Res.* 53 (01), 19–30.
- Miller, A., 2021. Ship model identification with genetic algorithm tuning. *Appl. Sci.* 11 (12).
- Nilsson, N.J., 2005. Decision trees. In: *Introduction To Machine Learning*. Stanford University, California, USA, pp. 73–87.
- Perera, L.P., Oliveira, P., Guedes Soares, C., 2015. System identification of nonlinear vessel steering. *J. Offshore Mech. Arct. Eng.* 137 (3).
- Shalev-Shwartz, S., Ben-David, S., 2014. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, Cambridge, UK.
- Society of Naval Architects and Marine Engineers (U.S.). Technical and Research Committee. Hydrodynamics Subcommittee, 1950. Nomenclature for treating the motion of a submerged body through a fluid: Report of the American towing tank conference. In: *Technical and Research Bulletin*, Society of Naval Architects and Marine Engineers, USA.
- Sutulo, S., Guedes Soares, C., 2014. An algorithm for offline identification of ship manoeuvring mathematical models from free-running tests. *Ocean Eng.* 79, 10–25.
- Suykens, J., Vandewalle, J., 1999. Least squares support vector machine classifiers. *Neural Process. Lett.* 9, 293–300.
- Tibshirani, R., 1996. Regression shrinkage and selection via the lasso. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 58 (1), 267–288.
- Ungarala, S., Dolence, E., Li, K., 2007. Constrained extended kalman filter for nonlinear state estimation. *IFAC Proc. Vol.* 40 (5), 63–68, 8th IFAC Symposium on Dynamics and Control of Process Systems.
- Vapnik, V., Levin, E., Lecun, Y., 1994. Measuring the VC-dimension of a learning machine. *Neural Comput.* 6, 851–876.
- Woo, J., Park, J., Yu, C., Kim, N., 2018. Dynamic model identification of unmanned surface vehicles using deep learning network. *Appl. Ocean Res.* 78, 123–133.
- Xu, P.-F., Han, C.-B., Cheng, H.-X., Cheng, C., Ge, T., 2022. A physics-informed neural network for the prediction of unmanned surface vehicle dynamics. *J. Mar. Sci. Eng.* 10 (2).
- Xu, H., Hassani, V., Guedes Soares, C., 2019. Uncertainty analysis of the hydrodynamic coefficients estimation of a nonlinear manoeuvring model based on planar motion mechanism tests. *Ocean Eng.* 173, 450–459.
- Xu, H., Hinostroza, M., Guedes Soares, C., 2018. Estimation of hydrodynamic coefficients of a nonlinear manoeuvring mathematical model with free-running ship model tests. *Int. J. Marit. Eng.* 160.
- Xu, H., Hinostroza, M., Wang, Z., Guedes Soares, C., 2020. Experimental investigation of shallow water effect on vessel steering model using system identification method. *Ocean Eng.* 199, 106940.
- Yoon, J., Kim, R., 2024. Identifying kinetic model parameters and implementing 3-DOF control for a dual-thruster USV: A case study using the VRX simulation environment. *ICINCO 2024*, In: *Proceedings of the 21st International Conference on Informatics in Control, Automation and Robotics*, vol. 1, pp. 368–375.
- Zhu, M., Hahn, A., Wen, Y.-Q., Bolles, A., 2017. Identification-based simplified model of large container ships using support vector machines and artificial bee colony algorithm. *Appl. Ocean Res.* 68, 249–261.